Domain	Specialist Deliverables	Strategic Value to Client Teams
Electronics & Hardware Systems	Power-path design, mixed-signal interfaces, embedded bring-up, platform stabilisation.	Provides a stable hardware foundation for full-cycle validation and rapid issue isolation.
Post-Silicon Validation (V&V)	PMU/PMIC, DC-DC, BOOST, AFE bring-up; PVT sweeps; silicon-to-model correlation.	Reduces risk of silicon re-spins and accelerates time-to-market.
Precision Analogue & Mixed-Signal	Low-noise front-ends, Delta-Sigma ADCs, sensor-chain optimisation, reference stability.	Ensures reliable analogue performance in noise-sensitive systems.
DCDC Power Conversion & Stability	Buck/boost validation, transient response, compensation tuning, switching-loss analysis.	Delivers stable power rails under dynamic load conditions.
Power Losses & Switching Physics	Conduction/switching loss modelling, reverse-recovery behaviour, thermal envelope analysis.	Improves lifetime, derating strategies, and thermal reliability.
Power Electronics & SiC/GaN	High-voltage gate-driver validation, protection behaviour, HF parasitic analysis.	Enables safe, efficient operation of wide-bandgap power stages.
GaN/SiC Device Characterisation	Static/dynamic curves, leakage, threshold drift, dv/dt & di/dt behaviour, SOA mapping.	Defines safe operating envelopes and identifies degradation trends.
Dynamic & Static Test-Jigs (incl. DPT)	DPT fixture design, loop-inductance control, parasitic minimisation, high-speed measurement integrity.	Provides reproducible switching-waveform capture for WBG validation.
Battery Support Devices / CMU / AFE / EIS	Cell-monitoring accuracy, noise/crosstalk, isolation interfaces, EIS-based diagnostics.	Strengthens pack-health assessment and measurement-chain reliability.
Sensor MEMS, ISFETs & 2D/3D Materials	MEMS inertial/pressure evaluation, ISFET electrochemical behaviour, thin-film/2D materials characterisation.	Supports next-generation sensing platforms with cross-domain insight.
Photonics Devices (High-Speed Optical)	Optical/IR/VCSEL validation, PAM4/TIA behaviour, wavelength/temperature drift, burst-mode TX/RX.	Ensures stable photonics performance under electrical and thermal stress.
Semiconductor Test Equipment (ATE-Class)	SMU/curve-trace pipelines, PXI/TestStand automation, HV/HF measurement-chain design.	Aligns lab-level validation with production-grade test methodologies.
Semiconductor Metrology & Physical Measurements	XRF, XPS, Raman, IR/UV-Vis, thin-film behaviour, contamination/oxidation analysis.	Links electrical anomalies to physical/material root causes.
MEMS Timing Devices & Frequency Control	MEMS oscillator/resonator validation, jitter/noise analysis, temperature-compensation behaviour, supply-sensitivity testing.	Ensures timing stability across environmental and electrical stress.
RF-Adjacent Validation (PVT & Power Behaviour)	TX/RX front-end PVT sweeps, burst-mode behaviour, supply droop, PLL lock dynamics, attach/detach stability.	Maintains RF subsystem coherence under dynamic load conditions.
Automation & Tooling	Python/MATLAB automation, PyVISA/Pytest frameworks, structured PVT sweeps, database-backed analysis.	Replaces manual bench work with scalable automated pipelines.